

036015

Finite Elements Method for Engineering Analysis

Homework Assignment 4 - Isoparametric Mapping

Winter 2026

Instructions

- Electronic submission of **typed** sheets only (PDF format through moodle).
- Submission file must be named: "*name surname.pdf*". No I.D number or homework number is required in the file name.
- Solution must contain all the relevant derivations, explanations, graphs and conclusions.
- All codes used must be added to the submission files.

Problem 1 - Macro Element Meshing

Consider a biquadratic macro element:

1. Draw a single macro element in its reference configuration and write the coordinates of each of its nodes. [5pt]
2. Write a code (in Matlab) that will subdivide the reference configuration into smaller linear triangular elements. [25pt]
The function that needs to be programmed:

```
[E,local_nodes]=innerMesh(mesh_refinement_factor,type)
```

type - is the type of the macro element: biquadratic (1) or quadratic triangular (2). (Can be skipped if not doing bonus question 1.3)

mesh_refinement_factor - number of subdivisions on ξ and η axes of the macro element. If this value is equal to k , in a biquadratic macro element for example, it should direct the creation of $n = 2 \cdot k^2$ triangular elements in the mesh. Note that $k \geq 2$.

local_nodes - $m \times 2$ matrix of the coordinates of each node in the reference configuration, where $m = (k + 1)^2$.

E - $n \times 3$ matrix of the node numbers of each linear triangular element in the meshed shape.

You may make changes to this function as long as you don't change the general concept.

3. Bonus [15pt]: Repeat this process for a quadratic triangular macro element.

Problem 2 - Isoparametric Mapping

Mesh a circular shape. The shape consists of six macro elements: two triangles and four quadrilaterals. If you didn't do problem 1.3, assume center element is a fifth quadrilateral. The shape:

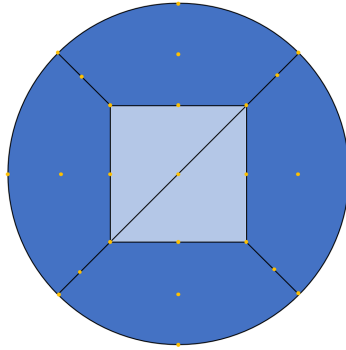


Figure 1: Triangular macro elements in light blue, and biquadratic macro elements in dark blue

The coordinates of the vertices of each macro element are given in the table below:

Macro Element	Vertices								
1	(-3,-3)	(3,3)	(-3,3)	(0,0)	(0,3)	(-3,0)			
2	(-3,-3)	(3,3)	(3,-3)	(0,0)	(3,0)	(0,-3)			
3	(-3,3)	(3,3)	(7,7)	(-7,7)	(0,3)	(5,5)	(0,10)	(-5,5)	(0,6)
4	(-3,-3)	(-3,3)	(-7,7)	(-7,-7)	(-3,0)	(-5,5)	(-10,0)	(-5,-5)	(-6,0)
5	(3,-3)	(-3,-3)	(-7,-7)	(7,-7)	(0,-3)	(-5,-5)	(0,-10)	(5,-5)	(0,-6)
6	(3,3)	(3,-3)	(7,-7)	(7,7)	(3,0)	(5,-5)	(10,0)	(5,5)	(6,0)



1. Define the isoparametric mapping that relates the natural coordinates of the reference configuration to the global coordinates of the actual shape. Do this by writing $N_i(\xi, \eta)$ explicitly. [10pt]

$$x_j = N_i(\xi_j, \eta_j)X_i \quad y_j = N_i(\xi_j, \eta_j)Y_i$$

Note that (X_i, Y_i) are the global coordinates of the macro element, and $N_i(\xi, \eta)$ are the basis functions of the macro element.

2. Write a code (in Matlab) that uses the isoparametric mapping to transform the smaller elements from the reference configuration to the global coordinate system. [40pt]
The function that needs to be programmed:

```
[E,N]=meshElement(macro,mesh_refinement_factor)
```

macro - is a 9×2 or a 6×2 matrix for a biquadratic or quadratic triangular macro element, respectively. This matrix holds the coordinates of the macro elements nodes in the global coordinate system. **Note:** make sure the order of the nodes matches the order of your N_i functions in their vector!

N - $m \times 2$ matrix of the coordinates of each node in the global coordinate system.

E - $n \times 3$ matrix of the node numbers of each linear triangular element in the meshed shape.

This function will be used in the final project where will you be asked to use macro elements to mesh a shape that is not necessarily a circle.

3. Use the two provided codes in the `mergeMesh_main.m` example to merge the meshes and to plot a meshed circle with labeled nodes and elements. Provide at least two pictures with different levels of refinement of the mesh. **Note:** make sure all macro elements have the same refinement factor. [10pt]
4. Briefly explain why finer meshes result in a rounder shape even though the macro elements are quadrilaterals. Is it really becoming more circular? Explain what shape is received. [10pt]
5. Bonus [8pt]: Calculate the Jacobian of all resulting elements to check bijectivity. Show their values on a histogram.

Good luck!